

Automated Web Scraping & Data Extraction Pipeline

Scrapes book listings from books.toscrape.com — a public sandbox site built specifically for scraping practice — and turns the raw HTML into structured CSV/JSON, plus an auto-generated summary report.

Features

- Custom headers (identifiable User-Agent)
- Rate limiting between page requests (`--delay`)
- Retry logic with exponential backoff on network/5xx errors
- HTML → structured fields: title, price, rating, availability
- Exports `output.csv` and `output.json`
- Prints + saves `summary_report.txt` (avg price, rating distribution, stock counts)

Setup

```
pip install -r requirements.txt
```

Run

```
python scraper.py --pages 5 --delay 1.5
```

- `--pages` : number of catalogue pages to scrape (default 5)
- `--delay` : seconds to wait between requests (default 1.5)

Output files

- `output.csv` — structured book data
- `output.json` — same data in JSON
- `summary_report.txt` — summary stats generated after each run

Submission

1. Push this folder to a public GitHub repo (include `scraper.py` , `requirements.txt` , and a real `output.csv` from a run against the live site).
2. Copy the repo's public URL.
3. Paste it into the "Submit deliverable" field on the RabTech Academy task page.